PROBABILITY 3: ASSIGNMENT 5

In the following problems $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X, Y denote r.v. on Ω.

Problem 1. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables such that $X_n \geq 0$ and $X_n \to X$ almost surely. If $\mathbb{E}[X_n] \to \mathbb{E}[X] < \infty$, then $\mathbb{E}[|X_n - X|] \to 0$.

Problem 2. In problem 1., if we drop the assumption $\mathbb{E}[X] < \infty$, does the conclusion $\mathbb{E}[|X_n - X|] \to 0$ still hold?

Problem 3. Suppose $X_n \to X$ a.s. Let μ_n and μ be the distribution of X_n and X respectively. Show that $\mu_n \xrightarrow{d} \mu$.

Problem 4. If X is r.v. such that $\mathbb{E}[|X|] < \infty$, show that there exist bounded random variables X_n such that $\mathbb{E}[|X_n - X|] \to 0$ as $n \to \infty$.

Problem 5. Find integrable random variables X_n, X for each of the following situations.

(1) $X_n \to X$ a.s. but $\mathbf{E}[X_n] \not\to \mathbf{E}[X]$.

(2) $X_n \to X$ a.s. and $\mathbf{E}[X_n] \to \mathbf{E}[X]$ but there is no dominating integrable random variable Y for the sequence $\{X_n\}$, i.e., there is no r.v. Y such that $|X_n| \leq Y$ and $\mathbb{E}[Y] < \infty$.